

Deep-Texture: A Lightweight Wearable Ring for Shape and Texture Rendering in Virtual Reality

Youjin Sung*

DongKyu Kwak†

Taeyeon Kim‡

Woontack Woo§

Sang Ho Yoon¶

KAIST
Daejeon, Republic of Korea

Figure 1: Illustration of our proposed solution Deep-Texture which is a foldable device that renders the shape and texture in Virtual Reality(VR). With our open-sourced Deep-Texture, you can easily download, print, and further develop these resources for your VR haptic projects.

ABSTRACT

In this paper, we suggest a foldable device that renders the shape and texture in Virtual Reality called Deep-Texture. We devised Deep-Texture to achieve effective haptic feedback with lightweight hardware by combining the basic sensations of shape and texture. By integrating the frequency change of linear resonant actuator and 1-bar mechanism, we propose a novel haptic interaction device for immersive VR experiences. During the pilot test, the results show that it enhances realism while keeping usability. By open-sourcing Deep-Texture, we aim to empower a wider range of users to engage with and benefit from haptic technology, ultimately lowering the hurdles.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction devices—Haptic devices;

1 INTRODUCTION

As the virtual reality (VR) experience is getting more realistic, researchers have become increasingly interested in providing tactile feedback. To render tactile information, FingerX [6], Pocopo [8], and Xrings [3] used a hand-held type. As wearability is not mainly considered, those previous examples are not suitable for the various hand interactions. Since they are limited to hand-held grasp types, the user's hand posture is always restricted to a grasping position. In terms of wearability, there is an attempt to design hardware as the foldable type. Teng et al. design a device that contacts only when an interaction occurs [5]. However, it gives single vibrotactile feedback which renders only a simple texture. In response, we developed Deep-Texture which is a foldable device that simultaneously renders

texture information with both tactile feedback and bar mechanism in real-time.

Deep-Texture is the simple form factor of the wearable type that renders shape and texture in real-time in VR. We combined LRA and 1-bar mechanism in real-time for basic texture and small geometric shapes. The contributions that we suggest are below. :

- For enhancing user immersion, we integrated vibration feedback with a 1-bar mechanism for rendering Deep-Texture.
- To improve user wearability, we designed our lightweight hardware as the foldable ring type.
- We published Deep-Texture as an open source¹ to allow users to build their own devices.

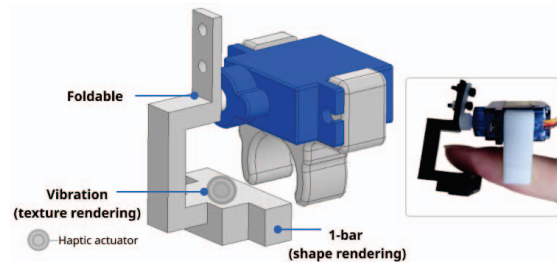


Figure 2: Core mechanism of Deep-Texture; foldable ring type, shape rendering with 1-bar, and texture rendering with vibration feedback

2 DEEP-TEXTURE

We developed the Deep-Texture, a wearable device that provides continuous shape and texture rendering on the index finger. Combining vibrotactile feedback and a 1-bar type rendering method further enhances the device's functionality, as it can render fine to rough textures. With its capabilities in a wearable form factor, users can

*672@kaist.ac.kr, These authors contributed equally to this work.

†kig160@kaist.ac.kr, These authors contributed equally to this work.

‡taeyeonkim@kaist.ac.kr

§wwoo@kaist.ac.kr

¶sangho@kaist.ac.kr, Corresponding author

¹<https://github.com/deeptexture/deeptexture>

experience a fully immersive VR experience. Moreover, the high wearability of the device allows enabling full flexion of all fingers even while wearing the device. The total weight of our hardware is about 16g which is reasonable compared to previous research FingerX(19g) [6].

Mechanism 1: Tactile Feedback In Deep-Texture, tactile feedback is used for rendering the texture of the object. High-frequency signals mimic fine surface details(i.e., material), while low-frequency signals simulate larger and more irregular features(i.e., mesh) [4]. In this project, we chose brick, cloth, and marble which are commonly used for texture classification for the basic three textures. We divided the frequency into three levels which are 30Hz, 100Hz, and 170Hz. Each level matches the three textures that we chose. (Brick-30Hz, Cloth-100Hz, and Marble=170Hz)

Mechanism 2: 1-bar linkage Our 1-bar mechanism rotates correspond to servomotor radius, so it is essential to clarify the precise angle at which the hand maintains contact with our device. Therefore, the angle range was obtained by actually wearing the devices which is about 80° . Although the contact point changes slightly due to the bar rotating, during pilot studies, we have confirmed that this does not have a significant impact.



Figure 3: Angle ranges of our devices is about 80° ($\pm 40^\circ$). 1-bar moves along with user's fingertip.

3 EXPLORING REALISM WITH DEEP-TEXTURE

We ran a pilot study to verify whether our solution enhanced the user's realism, comfort, and freedom. We set the 7-point Likert scale questionnaire from previous works [1, 2, 7].

3.1 Procedure

We recruited 5 participants ($M=26.8$, $SD=2.22$, 3 females, 2 males, all right-handed). They were equipped with our hardware and the VR HMD (Meta Quest Pro) and asked to move their index fingers to feel the edge of the 3D objects in all four conditions (Barehand (baseline), 1-bar (shape only), Vibration (texture only), and Multimodal (shape+texture)). Each condition has three textures (brick,

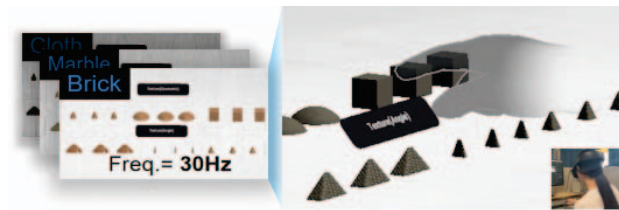


Figure 4: Participants can move their fingers from left to right to feel the texture with Deep-Texture. Each scene contains different textures (brick, cloth, and marble) with different frequencies.

cloth, marble), and each scene contains three shapes (sphere, hexahedron, tetrahedron). After 1 minute, they can freely interact with the virtual objects. The whole procedure took about 30 minutes.

Results Participants reported the highest realism ($M=5.2$, $SD=.68$) in our proposed Deep-Texture condition. We found statistical differences in the Realism sector ($p<.05$), and no significant differences in Comfort and Freedom sector. This means that it's not much of an inconvenience compared to barehand.

4 DISCUSSION & FUTURE WORKS

Shape and texture rendering By integrating vibration feedback with a 1-bar mechanism, our system can render Deep-Texture which includes small textures and geometric shapes in VR. For future works, we are planning to utilize visual illusion to enhance our multimodal system. This approach aims to create a more natural experience by aligning the visual and physical aspects of our system.



Figure 5: Example use case scenarios. A user interacts with the shapes and textures in the virtual scenes.

Open source In terms of accessibility, we anticipate that this development will significantly reduce the barriers for individuals using haptic devices. By open-sourcing our technology and providing hardware and software guidelines, we aim to empower a wider range of users, including those with needs and backgrounds, to readily engage with and benefit from haptic feedback experiences in various applications.

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